Gondwana Research 22 (2012) 447–460

Contents lists available at SciVerse ScienceDirect

Gondwana Research

journal homepage: www.elsevier.com/locate/gr

New paleomagnetic results of the early Permian in the Xainza area, Tibetan Plateau and their paleogeographical implications

Bo Ran [a,*], Chengshan Wang [b,*], Xixi Zhao [c], Yalin Li [b], Meng He [d], Lidong Zhu [e], Robert S. Coe [c]

[a] State Key Laboratory of Oil and Gas Reservoir Geology and Exploitation, Chengdu University of Technology, Chengdu, 610059, China
[b] State Key Laboratory of Geological Processes and Mineral Resources, China University of Geosciences (Beijing), Beijing, 100083, China
[c] Earth and Planetary Sciences Department, University of California, Santa Cruz, CA 95064, USA
[d] Liaoning Non-Ferrous Geological Exploration Institute, Shenyang 110013, China
[e] College of Sedimentary Geology, Chengdu University of Technology, Chengdu, 610059, China

ARTICLE INFO

Article history:
Received 26 February 2011
Received in revised form 20 November 2011
Accepted 21 November 2011
Available online 13 December 2011

Keywords:
Lhasa Terrane
Xainza area
Permian
Paleomagnetism
Paleogeography

ABSTRACT

The characteristic remanent magnetizations (ChRMs) of the late Paleozoic in the Lhasa Terrane reported previously were not widely used for paleogeographical reconstructions because the rocks have been obliterated significantly by multiphase orogeny. This study reports the paleomagnetic data of 131 samples collected from 17 sites in the Permian and the late Triassic rocks from the Zhakang section of the Xainza area, where the relative undeformed Paleozoic strata were preserved. Consequently, the ChRMs of high temperature with the different paleomagnetic directions were isolated from the Permian sandstones, limestones, dolomites and the late Triassic silicalites. Without the constraints with the spatial and temporal limitation of the chemical remagnetization derived from the fluids driven by the Neogene strike-slip faults and the Paleogene thrusting faults in the study area, the ChRMs carried by hematite in early Permian sandstones away the faults have potentially preserved a primary magnetization, which has a paleomagnetic direction of $D=46.6°$, $I=-31.2°$, $\kappa=69.0$, $\alpha95=9.3°$, $N=5$ sites after tilt correction, corresponding to a pole at 24.5°N, 219.3°E with $A95=7.8°$. The new corresponding paleomagnetic pole with the former Carboniferous poles shows the Lhasa Terrane drifted into the middle-lower latitude and isolated in the Paleo-Tethys Ocean rather than adjacent to the northern margin of Gondwanaland during the early Permian after being rifted from the Australian continent in the late Carboniferous.

© 2011 International Association for Gondwana Research. Published by Elsevier B.V. All rights reserved.

1. Introduction

The Lhasa Terrane, located in the southern Tibetan Plateau, is one of the most important elements for the paleogeography reconstruction of the Pangea since the late Paleozoic (e.g., Şengör, 1979; Yin and Harrison, 2000; Metcalfe, 2006; Aitchison et al., 2011). It is generally accepted that the Lhasa Terrane had rifted from Gondwana and then drifted northward before being finally amalgamated with the Qiangtang Terrane during the Cretaceous (e.g., Dewey et al., 1988; Yin and Harrison, 2000; Metcalfe, 2006), but recent studies indicate, on the basis of detrital zircon provenance study and new identification of several Paleozoic geological records, that the Lhasa Terrane should be the part of cratonic Australia in the late Paleozoic before being rifted in the late Carboniferous–early Permian (e.g., Zhu et al., 2009a, 2010, 2011a).

Paleomagnetic analysis is an effective approach to constrain the motive track of the Lhasa Terrane respective to eastern Gondwanaland (e.g., Zhao and Coe, 2007). Several recent plaeomagnetic studies have been conducted from various types of formations in Tibet and surrounding regions (e.g., Chen et al., 2011; Sun et al., 2011). However, because the primary magnetization in the rocks has generally been overprinted by multiphase deformation and thermal events in the orogen (e.g., Halim et al., 1998; Elmore et al., 2001; Torsvik et al., 2009), and the reliable quality from the late Paleozoic paleomagnetic data in the Lhasa Terrane could not satisfy with the lowest reliability criteria ($Q>2$) in the APWPs reconstructions (e.g., Ye and Li, 1987; Dong et al., 1990; van der Voo, 1990; Li et al., 2004). It is difficult to determine the paleogeographic affinity of the Lhasa Terrane with respect to eastern Gondwanaland in the late Paleozoic so that it is generally accepted by classical late Paleozoic Qiangtang–Lhasa–Himalaya–Greater India paleogeographic reconstructions (e.g., Scotese et al., 1999; Golonka and Ford, 2000; Yin and Harrison, 2000; Metcalfe, 2006).

In this study, we collected samples from Zhakang section in the Xainza area, where the late Paleozoic strata are well preserved (Ye and Li, 1987; Dong et al., 1990; Qu et al., 2003; Wang et al., 2003; Li et al., 2004; Yao et al., 2007), to determine the shielding area or rock styles with weak overprint of tectonic or thermal events (Figs. 1 and 2). The paleomagnetic data could have good potential preservation of primary ChRMs, which are expected to provide important constraints on the paleographical location of the Lhasa

* Corresponding authors. Fax: +86 10 82322171.
E-mail addresses: rabo1980@163.com (B. Ran), chshwang@cugb.edu.cn (C. Wang).

1342-937X/$ – see front matter © 2011 International Association for Gondwana Research. Published by Elsevier B.V. All rights reserved.
doi:10.1016/j.gr.2011.11.014

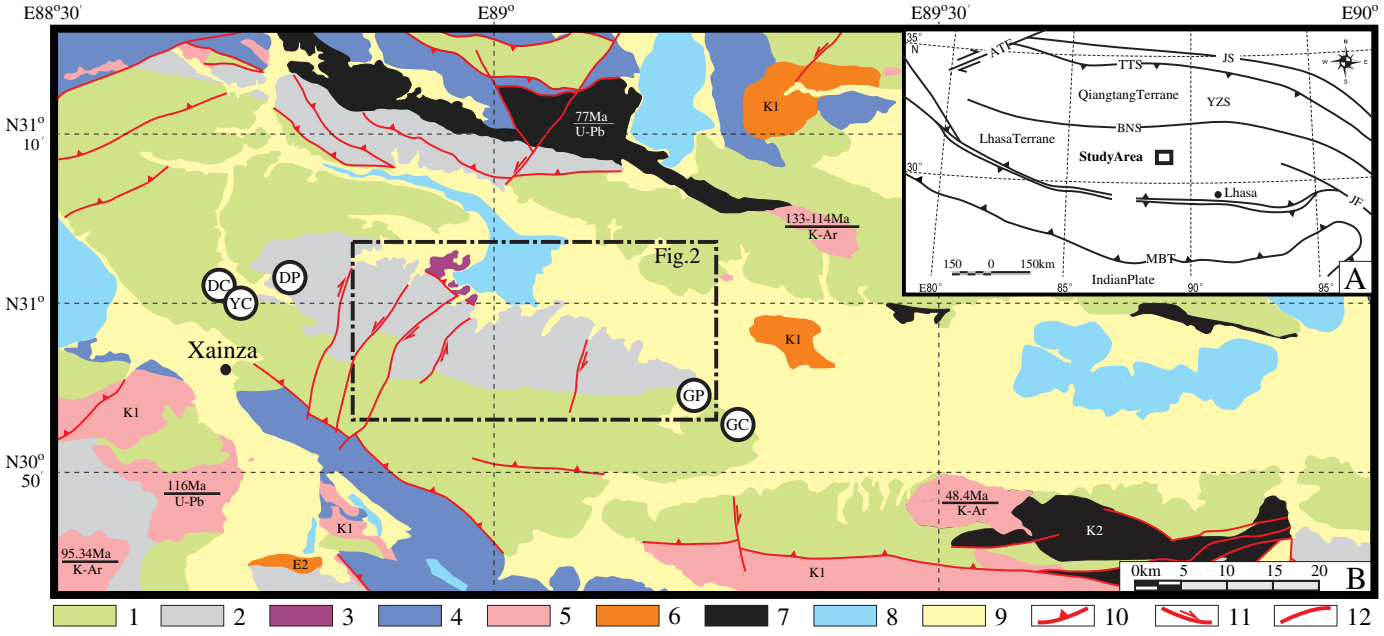


Fig. 1. (A) Sketch of structure in the Tibet Plateau (Zhu et al., 2006; Wang et al., 2008); (B) Simplified tectonic map showing the location of the Xainza area in the Lhasa Terrane (Qu et al., 2003; Wang et al., 2003). (1) before Permian; (2) Permian; (3) Upper Triassic; (4) Jurassic–Cretaceous; (5) lava; (6) granite; (7) ophiolite; (8) lake; (9) Quaternary; (10) thrust fault; (11) strike-slip fault; (12) unknown fault. TTS, Tanggula thrust system; JF, Jingsha fault; BNS, Bangong–Nujiang suture zone; YZS, Yarlung-Zangpo suture zone; MBT, Main Boundary Thrust; K1, Lower Cretaceous; K2, Upper Cretaceous; E2, Eocene; GC and GP, Carboniferous and Permian sampling sites from Guo (2009); DC and DP, Carboniferous and Permian sampling sites from Dong et al. (1990); YC, Carboniferous sampling sites from Ye and Li (1987).

Terrane during the late Paleozoic, and to reconstruct the bridge action for the stratigraphic and paleontological correlation between the Gondwanaland and Tethys realms in the Carboniferous–Permian time (e.g., Yao et al., 2007).

2. Geology and sampling

The Lhasa Terrane is situated between the Yarlung-Zangpo suture zone (YZS) and the Bangong-Nujiang suture zone (BNS) (Fig. 1A). The final closure of the BNS to the north of the Lhasa Terrane occurred in the Cretaceous (Allégre et al., 1984; Dewey et al., 1988; Yin and Harrison, 2000; Zhu et al., 2009b; Zhu et al., 2011b), and the closure of the YZS to the south of the Lhasa Terrane occurred during the late Cretaceous to Paleocene (Acton, 1999; Yin and Harrison, 2000; Aitchison and Davis, 2004; Chen et al., 2010; Dupont-Nivet et al., 2010; Najman et al., 2010; Sun et al., 2010).

Most complete and continuous Paleozoic sediments occur in the Paleo-Tethys region in the Xainza area, located at the center of the

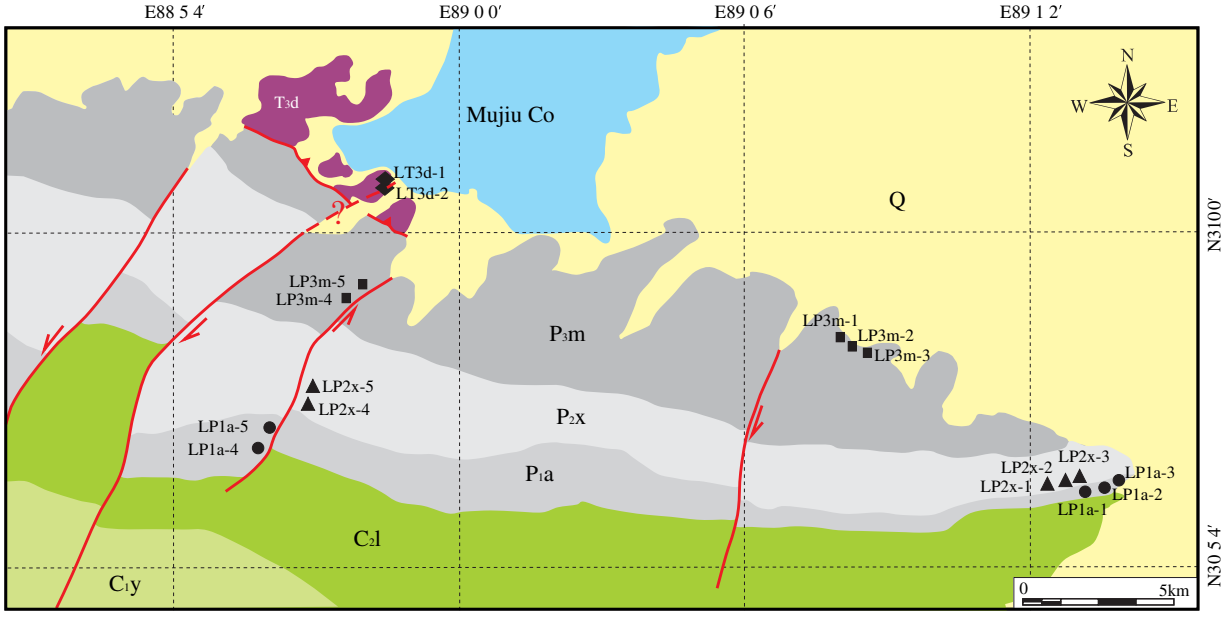


Fig. 2. Geological map of the study area (Qu et al., 2003; Wang et al., 2003). C_1y, Yongzhu Formation; C_2l, Laga Formation; P_1a, Angjie Formation; P_2x, Xiala Formation; P_3m, Mujiucuo Formation; T_3d, Duoburi Formation; Q, Quaternary; Triangle, Square, Diamond and Circle represent sampling locations.

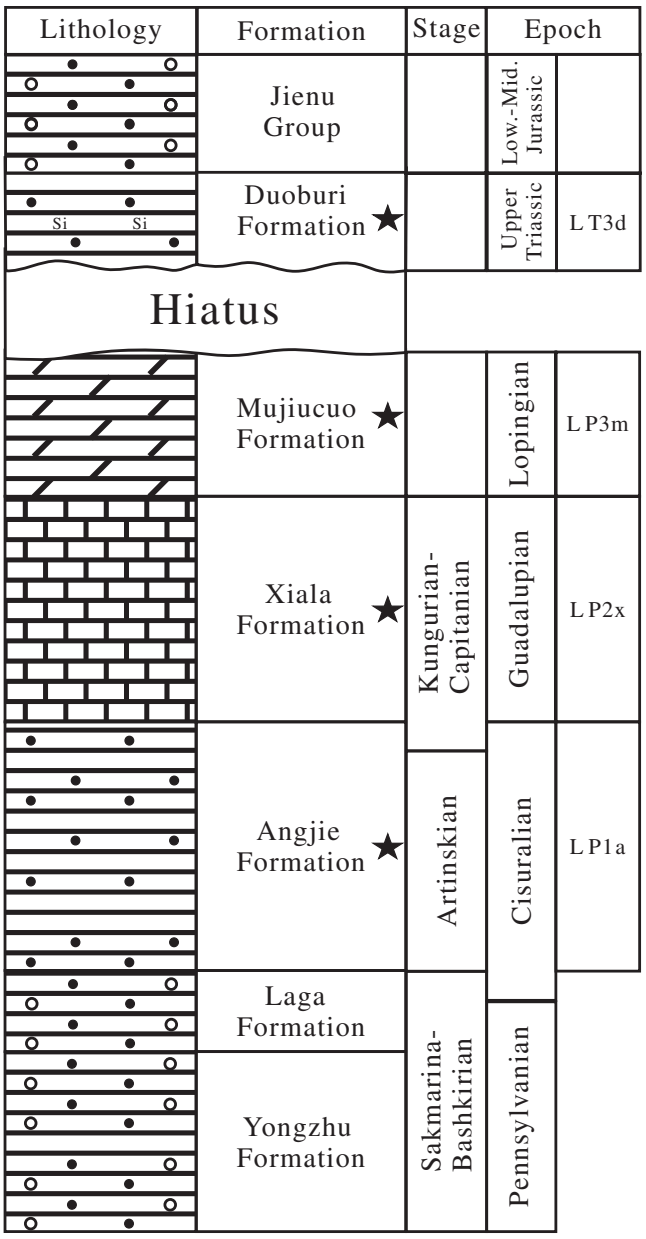


Fig. 3. Stratigraphic section showing the units sampled (BGMRX, 1993, 1997; Qu et al., 2003; Wang et al., 2003; Yao et al., 2007) and the approximate positions of the sample sites. Black stars represent the sampling strata.

Lhasa Terrane (Fig. 1A; BGMRX, 1993; Zheng et al., 2005; Yao et al., 2007). Based on previous geological data in the last century, our understanding of the biostratigraphy and lithostratigraphy of the study area has become more systematic (e.g., BGMRX, 1993; Jin, 2002; Qu et al., 2003; Wang et al., 2003; Yao et al., 2007). So the sampling stratigraphic scheme of this study is derived from these latest data (Fig. 3; e.g., Wang et al., 2003; Qu et al., 2003; Yao et al., 2007): from base to top, the Yongzhu Formation and the Laga Formation of Upper Carboniferous; the Angjie Formation of Lower Permian (Fig. 4A); the Xiala Formation of Middle Permian (Fig. 4B); the Mujiucuo Formation of Upper Permian (Fig. 4C); and the Duoburi Formation of Upper Triassic (Fig. 4D). In generally, there was absent to the early-middle Triassic in the Lhasa Terrane (Fig. 3; e.g., BGMRX, 1993; Zhu et al., 2009a, 2010).

The study area has been affected by strongly tectonic and magmatic events since the Mesozoic (e.g., BGMRX, 1993; Wu et al., 2003; Chung et al., 2005; Kapp et al., 2007a; Mo et al., 2007). The magmatic events in the Mesozoic are divided by three stages: the Zenong Group in the early Cretaceous (Qu et al., 2003; Wang et al., 2003; Zhu et al., 2008, 2009b); the Linzizong Group (LZZG) during the late Cretaceous to the Paleogene (e.g., Chung et al., 2005; Mo et al., 2007); the Wuyu Group in the Neogene (BGMRX, 1993, 1997; Qu et al., 2003; Wang et al., 2003). Influenced by the strong India-Lhasa collision, the study area mainly is controlled by two-stage Cenozoic tectonic actions: WNW-trending thrusting from the late Paleogene to the early Neogene during the hard-collision (Wu et al., 2003; Kapp et al., 2007a, 2007b); ENE-trending strike slipping in the Neogene in the post-collision (Coleman and Hodges, 1995; Blisniuk et al., 2001; Qu et al., 2003; Wang et al., 2003).

All 131 core samples of 17 sites were collected with a portable petro-powered drill and oriented by sun compass in Zhakang section, situated in the south of the Mujiu Co, Xainza country where the continuous Paleozoic strata are exposed with good access (Figs. 1 and 2). Some sites were oriented by both sun and magnetic compasses in order to evaluate any differences between local declinations and the present International Geomagnetic Reference Field in the sampling area.

3. Rock magnetic study

3.1. Techniques

Acquisition of isothermal remanent magnetization (IRM) and progressive thermal demagnetization of composite IRMs were carried out to identify coercivity and unblocking temperature spectra for five representative samples, one or two from each sampling strata. A direct current magnetic field was applied to acquire the IRM with a 2 G Enterprises Pulse Magnetizer with a maximum field of up to 2,700 mT. These samples were also subjected to progressive thermal demagnetization at temperatures of 100°, 200°, 300 °C, and then in intervals of 50°, 30° or 20° to 590 °C (or 690 °C) after applying a composite IRM following Lowrie (1990). DC magnetic fields of 2,700 mT; 400 mT; and 140 mT were applied to the sample Z-, Y- and X-axis, respectively. Remanent magnetizations for rock magnetic study were measured with a 2 G Enterprises cryogenic magnetometer (2 G–755) in the Paleomagnetism and Geochronology Laboratory (PGL) of the Institute of Geology and Geophysics, Chinese Academy of Sciences.

Alternating Field (AF) demagnetization was generally superior to thermal demagnetization at reducing the scattered in directions of natural remanent magnetization for samples. It was much better than thermal demagnetization of sister cores after removing secondary magnetization caused by lightning and almost as good at removing viscous components acquired over time at ambient temperature (Coe et al., 2005). In this study, combination with thermal and AF demagnetization was adopted to identify the high coercive parts (such as goethite or hematite) and low coercive parts (such as magnetite).

3.2. Results

Results of these experiments are shown in either IRM acquisition curves or demagnetization curves of the IRM (Fig. 5). The IRM acquisition experiments and thermal demagnetization of composite IRMs show that the limestone (including dolomite and silicalite) samples are dominated by magnetite. Indeed, the IRM of the sample usually reaches quasi-saturation at lower than 200 mT (Fig.5, sample LP2x1-5A), and the thermal demagnetization of the soft, medium, and hard components shows only one unblocking temperature of about 580 °C (Fig. 5). However, only using AF demagnetization, it was no complete to clean remanence for these samples with the relict over 10% (Fig. 6A, samples LT3d1-2B and LT3d1-3A), the thermal and AF demagnetization curves of the NRM show the low coercive parts

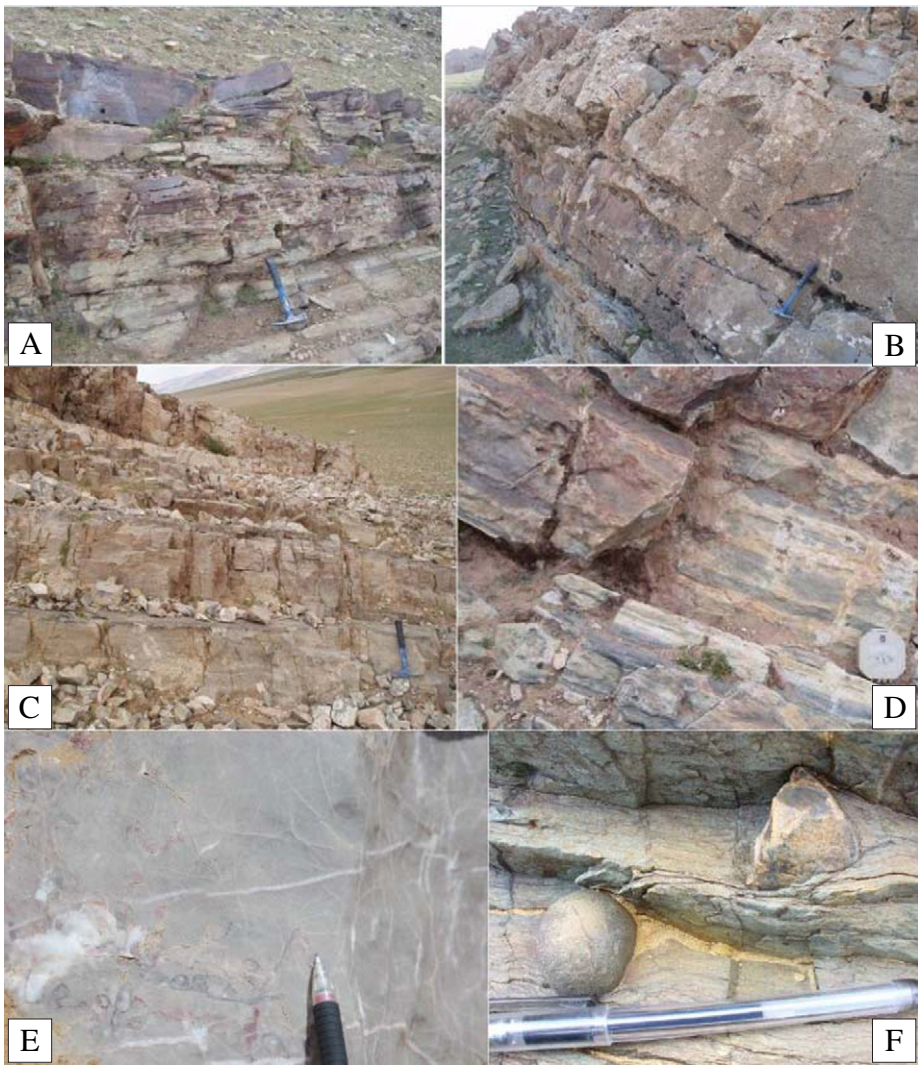


Fig. 4. Photographs showing the Permian and late Triassic strata in Zhakang section. (A) Angjie Formation (sandstones); (B) Xiala Formation (limestones); (C) Mujiucuo Formation (dolomites); (D) Duoburi Formation (silicalites); (E) calcite veins in the Xiala Formation; (F) dropstone in the Laga Formation in the Xainza area. The length of Hammer is 18 cm, compass 8 cm, pen 5.5 cm (E) and 10.5 cm (F).

could be completely cleaned by AF demagnetization after being heated till 150 °C (Fig. 6B).

For the sandstone, the acquisition behavior of IRM shows a trend of quasi-saturation in about 2,700 mT (Fig. 5, sample LP1a2-3A). Progressive thermal demagnetization of the composite IRMs reveals the hard component is unblocked by about 680 °C, meanwhile that the medium and soft components are unblocked by about 580–680 °C (Figs. 5 and 6), indicating that hematite is a main magnetic carrier with little influence of pyrrhotite or goethite (Figs. 5 and 6A, samples LP1a1-4A, LP1a1-7A, LP1a2-3A, and LP1a5-2A).

4. Paleomagnetic study

4.1. Techniques

Minicores, 2.5 cm diameter, were cut in the laboratory into 2.2–cm–long specimens. Magnetic measurements and demagnetization procedures were performed in the magnetically shielded room of the paleomagnetic laboratory of University of California, Santa Cruz (UCSC). In the laboratory, except for some limestone, dolomite and silicalite samples being subjected to hybrid thermal and AF demagnetization in order to obtain best demagnetization curve, the other samples (especially for sandstones) were progressively thermally demagnetized with a thermal demagnetizer. Most specimens were thermally demagnetized over 12 temperature steps up to 590 °C (or 680 °C) within a nearly zero field laboratory built furnace at UCSC. Demagnetization intervals were as large as 100 °C at the lower temperatures and as small as 20 °C at the higher temperatures, but in the lower temperatures (<150 °C) combining thermal and AF demagnetization, there were over five temperature steps. Progressive AF demagnetization was conducted using an AF demagnetizer 2 G–600 coupled to a 2 G–760 cryogenic magnetometer in steps of 10 or 20 mT. All remanence measurements were performed on a 2 G–760 cryogenic magnetometer of UCSC. Both demagnetizer and magnetometer are installed in a magnetically shielded space with the field inside minimized to<300 nT. During thermal demagnetization, sample orientation was successively inverted about the Z-axis in order to detect any systematic magnetization that could have resulted from the small ~10 nT residual magnetic field in the furnace.

Demagnetization results were plotted as orthogonal vectors diagrams (Zijderveld, 1967) and also as equal-area projections. Both AF and thermal cleaned paleomagnetic directions were determined using principal component analysis (Kirschvink, 1980). Site-mean (or sample-mean)

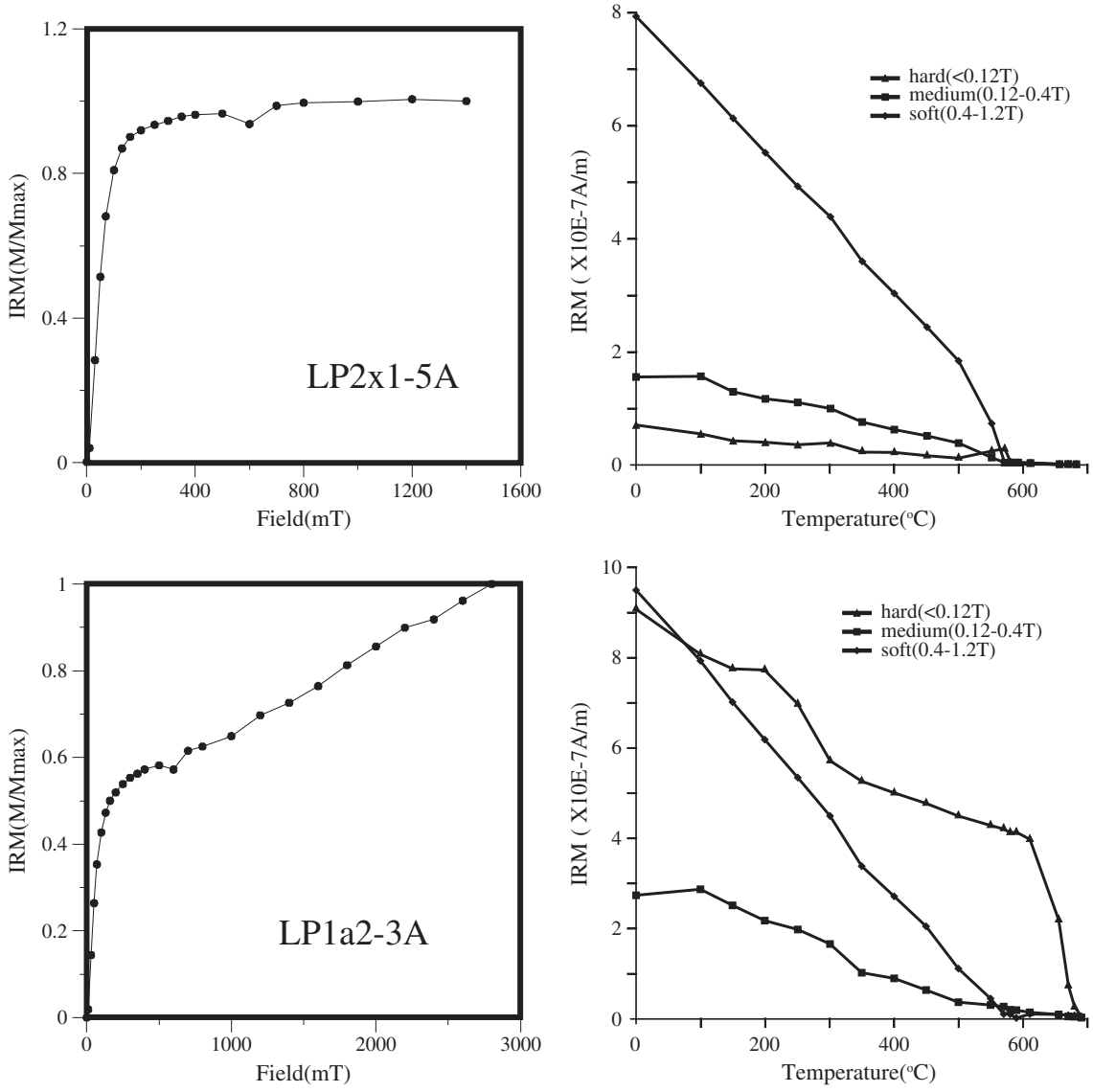


Fig. 5. Rock magnetic behaviour of samples LP2x1-5A and LP1a2-3A: (left) acquisition of isothermal remanent magnetization (IRM); (right) progressive thermal demagnetization curves of composite IRMs following Lowrie (1990).

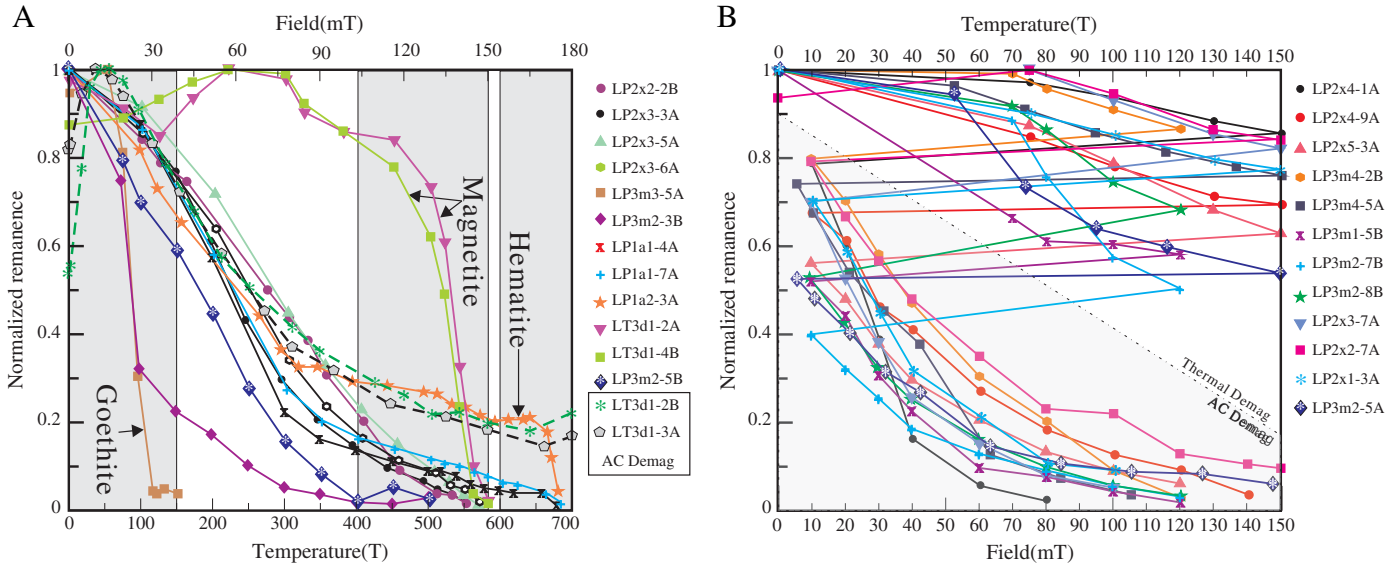


Fig. 6. Thermal (A) and AF (B) demagnetization curves of the NRM.

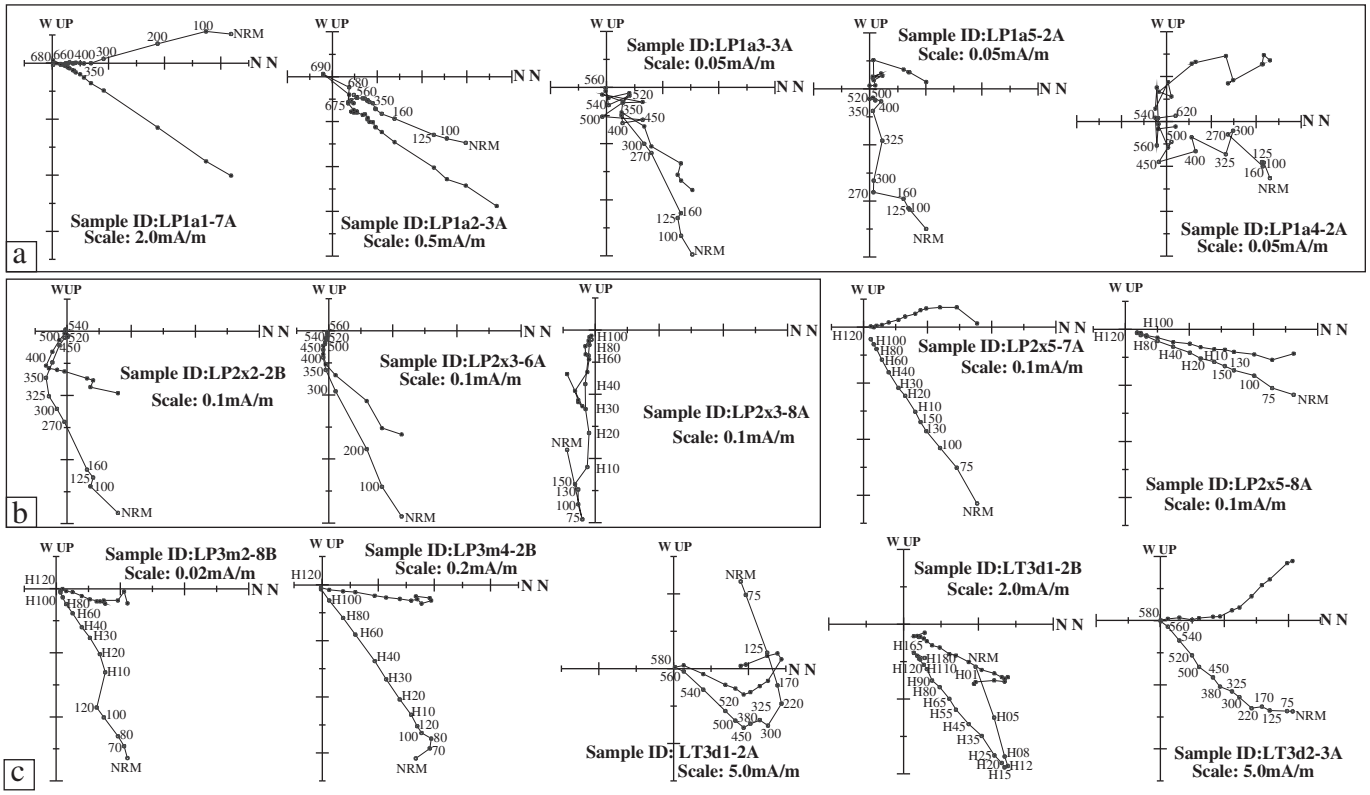


Fig. 7. Orthogonal vector plots of representative Permian and late Triassic samples. Directions are plotted *in situ*; open and solid circles represent vector endpoints projected onto the vertical and horizontal planes, respectively. H indicates AF (mT) demagnetization. Without H, only numbers indicate thermal (°C) demagnetization.

directions were calculated using Fisher (1953) statistics for directional data. All of the interpretations and data processing were carried out using Paleomagnetism Data Analysis Version 4.2 software (available from Randy Enkin's website).

4.2. Demagnetization results

The demagnetization results for 101 out of the 131 samples collected yield relatively simple demagnetization behaviors for both the Permian and late Triassic samples after the removal of recent viscous overprint components by temperature steps of 150 °C or 350 °C (Fig. 7).

4.2.1. Angjie Formation

The thermal demagnetization of the Angjie Formation sandstone samples is characterized by the appearance of the low- and high-temperature components in different directions (Fig. 7a). The low-temperature component (hereafter designated component A) was generally removed by 350 °C. This component has a NNE declination and a shallow, upward inclination *in situ* for site LP1a1, but a NNE declination and a steep, downward inclination *in situ* for sites LP1a2 and LP1a3. The high-temperature component (designated component B) was usually unblocked at 680–690 °C. This component reveals a NNE declination and shallow, upward directions *in situ* (Fig. 7a). Meanwhile, the high-temperature component B in sites LP1a3, LP1a4 and LP1a5 were disregarded due to chaotic demagnetization behavior (Fig. 7a, samples LP1a4-2A and LP1a 5-2A).

4.2.2. Xiala Formation

The thermal demagnetization of the Xiala Formation limestone samples is characterized by the appearance of two different direction components compared with those of the Angjie Formation.

The low-temperature component A in these sites was generally removed below 350 °C, however, remanence for these samples could be completely removed through thermal demagnetization (<150 °C) and AF (Fig. 6B), which show the magnetic carrier may be goethite. This component has a NNE declination and medium, downward inclination in *in-situ* coordinate. The high-temperature component B was usually unblocked by 540–560 °C, or 100 mT, and reveals a NEE declination and medium, downward directions for sites LP2x1, LP2x2 and LP2x3 *in situ* (Fig. 7b), however, the

Fig. 8. Equal-area projection of component A and B from Permian and late Triassic samples before and after bedding corrected. (A) component A from sites LP2x1, LP2x2 and LP2x3 *in situ* ($Dg = 28.9°$, $Ig = 63.9°$, $k = 76.5$, $\alpha95 = 14.2°$; $Ds = 356.5°$, $Is = 34.6°$, $k = 28.1$, $\alpha95 = 23.7°$; N = 3 sites; K = 22.1%, $D = 16.7°$, $I = 58.8°$, $k = 94.22$, $\alpha95 = 12.8°$); (B) component A from samples of sites LP2x4 and LP2x5 and LP3m1, LP3m2, LP3m3, LP3m4, LP3m5 and LT3d1-2 *in situ*; (C) and (D) component B *in situ* and bedding corrected from sites LP2x4, LP2x5, LP3m1, LP3m2, LP3m3, LP3m4, LP3m5 and LT3d1-2 along the faults except for the sandstone sites LP1a4 and LP1a5; (C′) and (D′) component B *in situ* and bedding corrected from sites LP2x4, LP2x5, LP3m1, LP3m2, LP3m3, LP3m4, LP3m5 and LT3d1-2, with component A from sites LP2x1, LP2x2 and LP2x3 *in situ* (optimal untilting at 14.5 ± 6.2% unfolding, $D = 7.6°$, $I = 52.6°$, $k = 44.3$, $\alpha95 = 6.9°$); (E) and (F) component B from limestone sites LP2x1, LP2x2 and LP2x3 away the faults *in situ* and bedding corrected; (G) and (H) component B from sandstone sites away the faults *in situ* and bedding corrected; (I) and (J) equal area net showing site-mean directions *in situ* and tilt-corrected coordinates in the study area including two sites of this study and three sites of Guo (2009). Solid/open circles represent downward/upward dipping directions with 95 per cent confidence limits. The solid square represents the mean direction. The solid star represents the mean direction in the PGF (Torsvik et al., 2008). *Dg*/*Ds*: declination *in situ*/unfolding, *Ig*/*Is*: inclination *in situ*/bedding-corrected, *k*: precision parameter, *α95*: 95% confidence angle, N, number of sites accepted.

component B, in sites LP2x4 and LP2x5, was usually unblocked by 100–120 mT, and reveals NNE and medium, downward directions *in situ* (Fig. 7c, samples LP2x5-7A and LP2x5-8A).

4.2.3. Mujiucuo formation

The combination with thermal and AF demagnetization of the dolomite samples of the Mujiucuo Formation is also characterized by

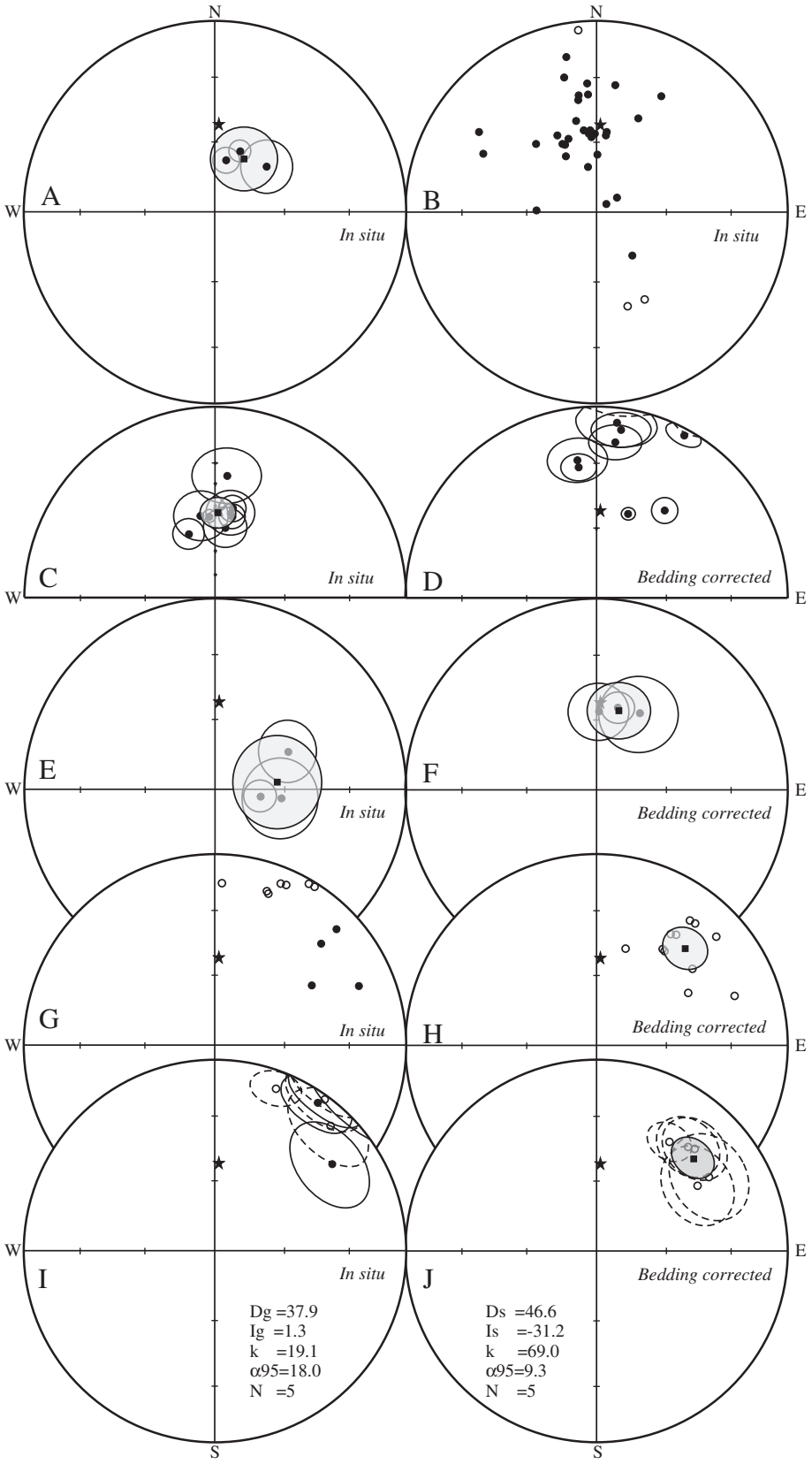


Table 1
Summary of paleomagnetic results of the ChRM from the Xainza area, the Lhasa Terrane.

Relationship with the fault	Sites	GPS coordinates		Bedding attitude		Mean direction							
		Latitude (° N)	Longitude (° E)	Dip direction (°)	Dip (°)	N/No	Geographic coordinates				Stratigraphic coordinates			
							D geo (°)	*I geo* (°)	*κ*	*α95* (°)	*D str* (°)	*I str* (°)	*κ*	*α95* (°)
Along the fault	LP1a4	30°55′54.6″	88°56′25.2″	322.1	26.0	6/6	–	–	–	–	–	–	–	–
	LP1a5	30°55′51.1″	88°56′26.8″	336.9	37.5	5/5	–	–	–	–	–	–	–	–
	LP2x4	30°56′48.1″	88°56′32.2″	289.1	35.5	6/7	337.3	60.1	104.0	6.6	352.3	31.1	104.0	6.6
	LP2x5	30°56′48.2″	88°56′30.0″	266.1	26.5	5/7	349.9	53.9	49.3	11.0	352.0	27.5	49.3	11.0
	LP3m1	30°58′03.2″	89°08′47.8″	281.1	28	6/7	5.4	35.6	26.3	13.3	6.5	7.9	27.1	13.1
	LP3m2	30°58′01.1″	89°08′48.0″	275.1	41.0	9/10	10.1	52.5	28.6	9.8	8.2	11.6	28.6	9.8
	LP3m3	30°57′57.7″	89°08′45.2″	275.1	41.0	6/11	8.0	59.3	57.6	8.9	6.6	18.3	57.6	8.9
	LP3m4	30°59′13.3″	88°58′08.8″	356.1	26	6/8	11.5	54.0	149.3	5.5	37.2	41.2	149.3	5.5
	LP3m5	30°59′13.5″	88°58′10.8″	357.1	19	7/9	355.4	54.5	500.5	2.7	20.3	50.8	500.5	2.7
	LT3d1-2	31°00′58.4″	88°58′24.1″	324.0	55	11/12	10.8	50.4	65.1	5.7	28.1	3.5	65.1	5.7
	Sites mean					(8/10)	1.8	53.1	61.2	7.1	10.7	24.6	13.9	15.4
Away the fault	LP2x1	30°55′33.5″	89°12′46.6″	254.1	41.0	13/13	98.2	70.6	35.0	7.1	14.4	53.2	35.0	7.1
	LP2x2	30°55′35.9″	89°12′47.1″	254.1	41.0	8/9	97.1	61.5	11.7	16.9	29.2	51.8	11.7	16.9
	LP2x3	30°55′36.1″	89°12′47.5″	210.1	39.0	3/5	62.4	54.4	95.3	12.7	1.5	55.9	93.8	12.8
	Sites mean					(3/3)	82.9	63.2	40.8	19.6	15.5	54.2	92.7	12.9
	Samples mean					24/27	85.2	66.7	21.2	6.6	16.3	51.0	23.5	6.2
	Samples mean (66.4% unfolding)					24/27					30.0	60.7	24.3	6.1
	LP1a1	30°55′24.4″	89°12′52.6″	242.1	37.0	7/10	21.5	−10.0	48.0	8.8	33.6	−31.7	48.0	8.8
	LP1a2	30°55′26.2″	89°12′53.1″	284.1	69.0	4/5	54.3	24.2	28.8	17.4	57.0	−30.3	28.5	17.5
	LP1a3	30°55′28.6″	89°12′50.0″	287.6	40.5	0/7	–	–	–	–	–	–	–	–
	Sites mean					11/22	32.7	2.3	9.5	15.6	42.1	−31.6	26.1	9.1
	SP01	30°54′	89°18′	270.0	40.0	6/7	43.3	−11.3	16.9	16	57.9	−37.4	16.9	16
	SP02	30°54′	89°18′	268.0	44.0	6/6	35.3	5.9	28.6	12.6	41.5	−28.3	28.6	12.6
	SP03	30°54′	89°18′	260.0	38.0	5/8	36.4	−1.8	19.6	13.2	44.0	−26.8	19.6	13.2
	Sites Mean [Including the Angjie Formation data of Guo (2009)]					5/6	37.9	1.3	19.1	18.0	46.6	−31.2	69.0	9.3
Adding the Carboniferous data of Guo (2009)														
Yongzhu Fm.	SC01	30°50′	89°23′	295.0	89.0	6/8	271.0	−1.9	27.8	12.9	296.7	−24	27.8	12.9
	SC02	30°50′	89°23′	235.0	58.0	4/8	191.1	−17.3	10.5	31.4	193.8	23.8	10.5	31.4
	SC04	30°50′	89°23′	263.0	42.0	6/8	300.2	76.6	5.3	32.0	191.7	54.8	5.3	32.0
	SC05	30°48′	89°20′	222.0	28.0	4/7	114.5	9.3	38.4	15.0	110.5	35.8	38.4	15.0
	SC07	30°52′	89°25′	248.0	37.0	4/8	127.4	19.0	5.7	31.1	139.8	43.6	5.7	31.1
	Sites mean					5/5	127.4	19.0	2.5	21.5	139.8	43.6	4.9	13.6

D: declination; *I*: inclination; *k*: precision parameter; *α95*: 95% confidence angle; N/No, number of samples (or sites, where in parentheses) accepted/studied.
Angjie formation's fold test:

(1) Watson and Enkin's Test (1993): the optimum concentration is achieved at 103.2 ± 33.8% unfolding;
(2) McElhiny's Test (1964): N = 5, *ks*/*kg* = 3.623 > F[8,8] = 3.44 of the grouping of the ChRM, at 95% confidence limits, indicative of a positive fold test.
(3) Enkin's Test (2003): N = 5, optimal untilting at 91.2 ± 61.8% unfolding, indicate of a positive fold test.
(4) McFadden's Test (1990): N = 5, in-situ $\xi_2 = 2.863$, tilt-corrected $\xi_2 = 323$, statistical threshold $\xi = 2.609$ at 95% confidence level, indicate of a positive fold test.

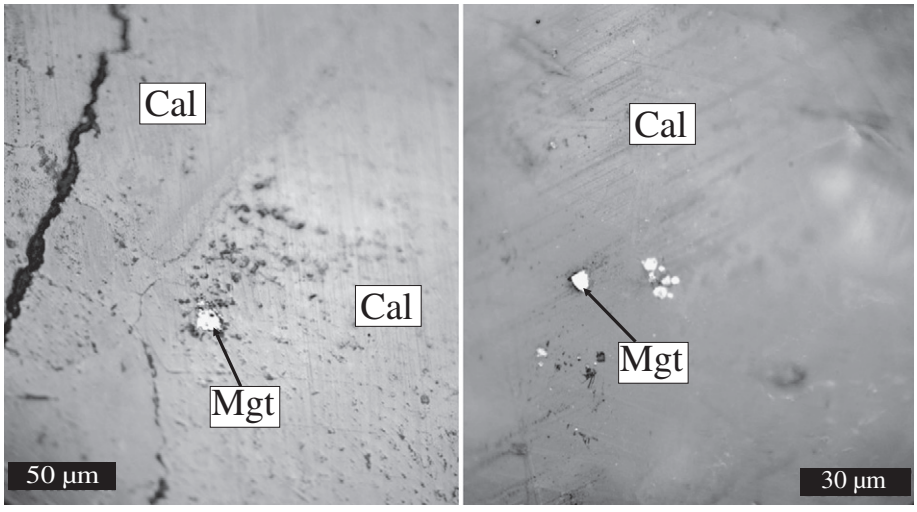


Fig. 9. Polished thins photographs of samples from the gray limestone of the Xiala Formation (sample LP2x4-2B). Mgt, magnetite; Cal, calcite. Euhedral magnetite filling one calcite grain.

the appearance of low- (A) and high-temperature (B) components with different directions (Fig. 7c). The component A was generally removed by 120 °C demagnetization, which is so scattered that it may be of viscous origin. The component B was usually unblocked by 120 mT, and reveals a NNE declination and shallow, downward inclinations *in situ* coordinate (Fig. 7c, samples LP32-8B and LP3m4-2B).

4.2.4. Duoburi formation

The combination with thermal and AF demagnetization of the Duoburi Formation silicalite samples is characterized by the appearance of two obviously different components (Fig. 7c). The component A was generally removed by below 380 °C or 15 mT, which is so scattered and disregarded. The high-temperature component B was usually unblocked by 580 °C or 120 mT, and reveals a NNE and shallow, downward directions *in situ* (Fig. 7c, samples LT3d1-2B and LT3d2-3A).

4.3. Summary of paleomagnetic results

The component A can be isolated from the Permian limestones (sites LP2x1, LP2x2 and LP2x3), which was usually removed by about 350 °C (Fig. 7). Site-mean of the component A does not pass the fold test (McFadden, 1990) indicating it may be a recent overprint. Furthermore, it is directed to the 28.9° with about 63.9° downward inclination *in situ*, but the component A of the other sites is scattered so that it is difficult to interpret (Fig. 8A, B).

The component B can be isolated from most samples of the Permian and late Triassic sandstones, limestones, dolomites and silicalites, except for the component B of sites LP1a4 and LP1a5 sandstones being destroyed. Then these high-temperature (B) components were divided into three different parts:

(1) Part ①: from sampling sites LP2x4, LP2x5, LP3m1, LP3m2, LP3m3, LP3m4, LP3m5, LT3d1 and LT3d2, it is dominated by normal polarity and has an *in situ* site-mean direction of $D=1.8°$, $I=53.1°$, $\kappa=61.2°$, $\alpha95=7.1°$ and a bedding-corrected one of $D=10.7°$, $I=24.6°$, $\kappa=13.9°$, $\alpha95=15.4°$ (Fig. 8C, D and Table 1). Because of the ratio $ks/kg=0.227<F[14, 14]$, it is indicative of a negative fold test. Then, because the direction of the component A from the sites LP2x1, LP2x2 and LP2x3 is similar with of the component B *in situ*, the composite directions is $D=7.6°$, $I=56.5°$, $\kappa=41.4°$, $\alpha95=7.2°$, which is close to the present-day geomagnetic field (PGF) direction ($D=2.0°$, $I=52.0°$, $\alpha95=2.7°$) *in situ* (Torsvik et al., 2008). Hereafter, this remagnetization event of the component B is called I–Remagnetization (I–Remag).

(2) Part ②: from sampling sites LP2x1, LP2x2 and LP2x3, it is dominated by normal polarity and has an *in situ* site-mean direction of $D=82.9°$, $I=63.2°$, $\kappa=40.8°$, $\alpha95=19.6°$, and an unfolding one of $D=15.5°$, $I=54.2°$, $\kappa=92.7°$, $\alpha95=12.9°$; sample-mean direction of $D=85.2°$, $I=66.7°$, $\kappa=21.2°$, $\alpha95=6.6°$*in situ*, and a bedding-corrected of $D=16.3°$, $I=51.0°$, $\kappa=23.5°$, $\alpha95=6.2°$, with the maximum Fisher's precision parameter $\kappa=24.3°$ in the 66.4% unfolding. These directions after bedding-corrected are also close to the PGF (Fig. 8E, F and Table 1). So it is obvious that the component B was remagnetizated being called II–Remagnetization (II–Remag).

(3) Part ③: from sampling sites LP1a1 and LP1a2, sample-mean directions were calculated for only two sampling sites because the component B of site LP1a3 is isolated from one sample (Fig. 7, sample LP1a3-3A) and the other samples in this site could not isolate the good component B. The reason may be correlated with the coarse grains of the sampling sandstone. It has an *in situ* mean direction of $D=32.7°$, $I=2.3°$, $\kappa=9.5°$, $\alpha95=15.6°$ and a bedding-corrected of $D=42.1°$, $I=-31.6°$, $\kappa=26.1°$, $\alpha95=9.1°$ for these samples (Fig. 8G, H and Table 1). Although it is impossible to subject to the fold test of two sites mean, it is a good chance to subject to the fold test of five sites after adding three good sites clustering directions of the Angjie Formation in the same section (Fig. 1 and Table 1; Guo, 2009). These samples of Guo (2009) were collected in sites with N 30°54′, E 89°18′ and most specimens were thermally demagnetized over 15 temperature steps up to 680 °C and measured with a 2 G Enterprises cryogenic magnetometer 755R within a ASC TD-48 furnace in the PGL. So five sites collected provide a site-mean direction for the component B of $D=37.9°$, $I=1.3°$ ($\alpha95=18.0°$) before and of $D=46.6°$, $I=-31.2°$ ($\alpha95=9.3°$) after tilt correction (Fig. 8I, J), which is indicative of a positive fold test with obvious improvement of data grouping following tilt correction, and at 95% confidence limits, yielding the ratio $ks/kg=3.623>F[8,8]$ of the grouping of the ChRM (McElhinny, 1964), and with optimal untilting at $91.2\pm61.8\%$ (Table 1; Enkin, 2003). Following application of the McFadden (1990) fold test, the result is positive with in-situ $\xi_2=2.863$, tilt-corrected $\xi_2=1.323$, and statistically critical $\xi=2.609$ at the 95% confidence level. This indicates that the component B defined from five sites is prefolding and primary with the oldest hiatus in the middle – early Triassic.

5. Discussion

5.1. Remagnetizaion in the Xainza area

The directions of paleomagnetic ChRM of I–Remag and component A of II–Remag are close to the PGF *in situ*. Several possible mechanisms can cause this remagnetization. Firstly, surface weathering should be excluded because all magnetic carriers (magnetite and goethite) recorded extremely well-clustered directions ($\alpha95=7.2°$), and there are different directions of magnetite components of the limestones in the different positions, which are hard to acquire by weathering processes in the local area (e.g., Park et al., 2003). Secondly, a thermoviscous remanent magnetization (TVRM) origin for the remagnetization can be proposed. The maximum unblocking temperature drops related to the strike-slip fault distance: Along these strike-slip faults, the maximum temperature was about 580 °C (Fig. 7a); but away these faults, the maximum temperature decreased to 400–450 °C (Fig. 6A, samples LP3m2-3B and LP3m2-5B), then the unblocking temperature of the component A dropped below 350 °C far away the faults (Fig. 7b). Meanwhile there are only Quaternary sediments during the Neogene in the sampling area (Fig. 1), so that the effects from the palaeotemperatures, significantly lower than temperatures required to have activated the remagnetization for magnetite with an unblocking temperature of 500–580 °C, should be faint.

The direction of magnetite components with maximum unblocking temperatures of 580 °C are consistent with the direction of the goethite component A, which implies that two components were acquired by a single process at almost the same time. Euhedral magnetite grains distributed into one single calcite grain are observed in gray limestone sample (sample LP2x4-2B) (Fig. 9), which these features imply that most magnetic grains generally have authigenic characteristic rather than a detrital nature (e.g., Park et al., 2003). So the chemical remanent magnetization is most likely to be the major mechanism of remagnetization in the study area. In generally, many studies have reported the chemical remanent to be connected to the tectonically driven external hydrotherm fluid migration (e.g., Oliver, 1986; Park et al., 2003; Enkin et al., 2007). Based on maximum unblocking temperature of remagnetization components drops with the distance to the orogenic fluids driven by strike-slip faults and euhedral magnetic grains (Figs. 1 and 8), it is interpreted that a fluid process driven by the Neogene strike-slip faults is most likely to be responsible for acquisition of I–Remag.

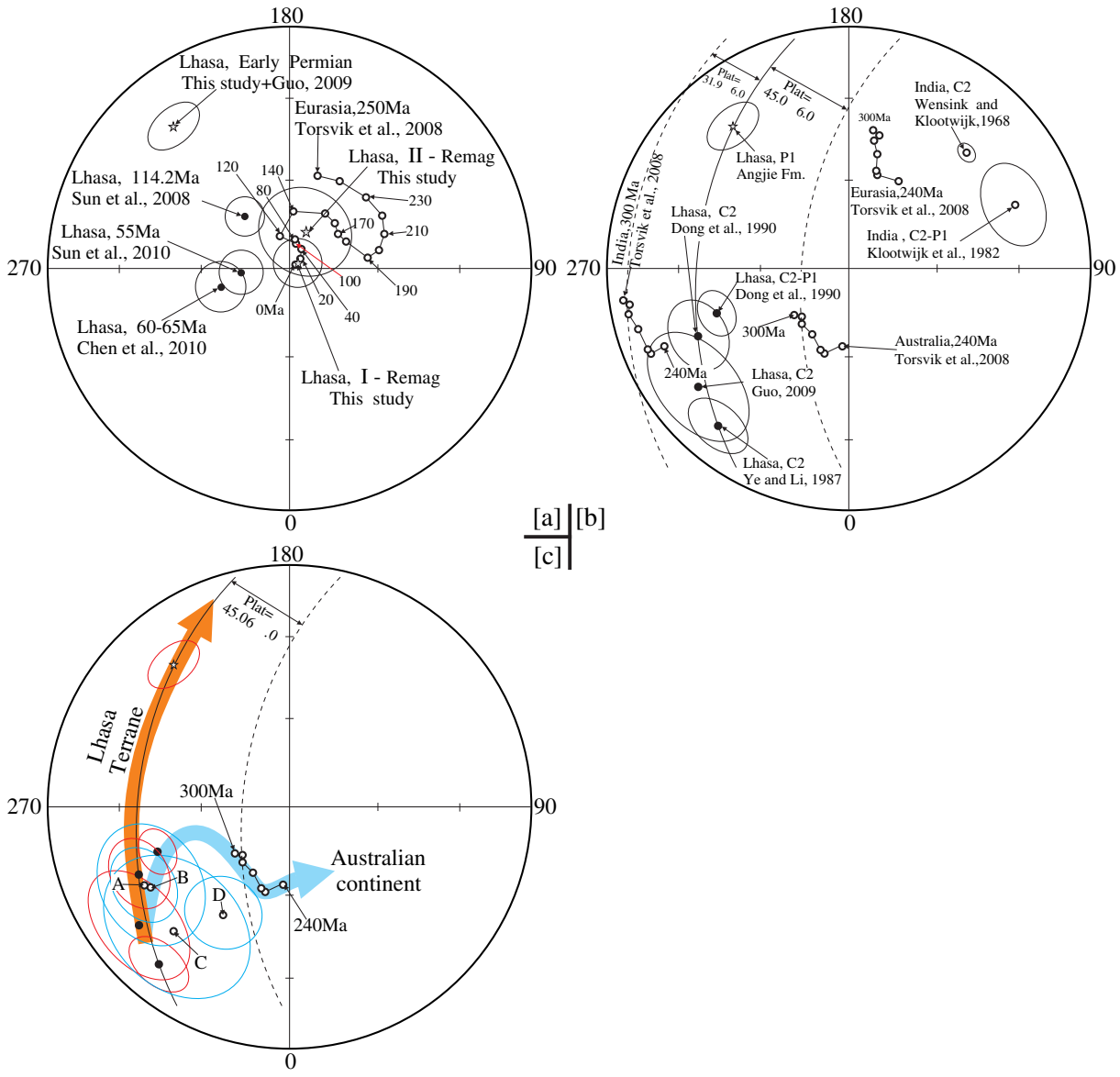


Fig. 10. a) Equal-area projections showing comparisons between paleomagnetic poles from the Lhasa Terrane in the former studies and the new paleomagnetic poles from Permian and Upper Triassic in Zhakang section of the Xainza area, the Lhasa Terraneand the master APW path for the Eurasian Plate since 250 Ma (Torsvik et al., 2008). b) Equal-area projection of the former Carboniferous and early Permian poles of the Lhasa Terrane and adjacent continents including cratonic India, Australian continent and Eurasian Plate. See Table 2 for values and references. dotted line: small-circle running through the 290 Ma reference pole of India (Australia) centered on our site locations; solid lines: small-circles passing through our site locations. c) Equal-area projection of the former Carboniferous and early Permian poles of the Lhasa Terrane and Australian continent. See Table 2 for values and references. Paleopoles are plotted onto the northern Hemisphere with 95% confidence limits. Gray stars: the new paleomagnetic poles of the Lhasa Terrane in this study; solid circles: the former paleomagnetic poles of the Lhasa Terrane; open circles: the paleomagnetic poles from the Indian continent, Australian continent and Eurasian Plate.

The paleomagnetic pole position calculated from the remagnetized ChRM component of II–Remag from the sampling sites LP2x1, LP2x2 and LP2x3 away the strike-slip faults is at 76.4° N, 158.4° E ($A95 = 15.2°$) (76.1° N, 171.7° E, $A95 = 6.9°$ from sample-mean) after bedding-corrected is close to the PGF (Fig. 10a; Torsvik et al., 2008), which indicates II–Remag is not the result of the remagnetization similar to I–Remag with the folding age before the Neogene (Fig. 11; Wang et al., 2003; Qu et al., 2003; Wu et al., 2008). In the three sampling sites, the differences in bedding attitudes between the different sites is small and thus it is difficult to obtain a meaningful fold test (Table 1). The progressive unfolding process reveals that the sample-mean directions give a maximal concentration of direction at about 66.4% unfolding (with uncertainties ranging from 59.7 to 72.7%) ($D = 30.0°$, $I = 60.7°$, $\alpha95 = 6.1°$; Watson and Enkin, 1993), which is statistically indistinguishable from that of 100% unfolding ($D = 16.3°$, $I = 51.0°$, $\alpha95 = 6.2°$). The abnormal paleolatitude $41.7 \pm 8.1°$ N of the sample-mean direction much higher than the present latitude of the Lhasa Terrane indicates the maximal concentration of direction is not statistical meaning, however, implies the characteristics of ChRM may be acquired in the process of pre-Miocene folding (Fig. 11). There are two possibilities for II–Remag in the Paleogene: (1) thermal effects of plutons or lavas, (2) the fluids driven by the faults. Based on the distribution of the magmas in the study area, there are Paleogene magmas (48.4 Ma, K-Ar age) and late Paleogene magmas nearby the sampling area excluding the extensive Cretaceous magmas in the study area (Fig. 1; Wang et al., 2003; Qu et al., 2003). However the distance between sampling sites (LP2x1, LP2x2 and LP2x3) and the Paleogene magmas excess 10 km (Fig. 1), which contradicts short-distance effects from hydrothermal fluids (e.g., Katz et al., 1998). So that the first thermal effects of the Paleogene magmas are impossible, and the second remagnetization mechanism of tectonic fluid is viable, because (1) WNW-trending thrusting faults mainly occurred in the

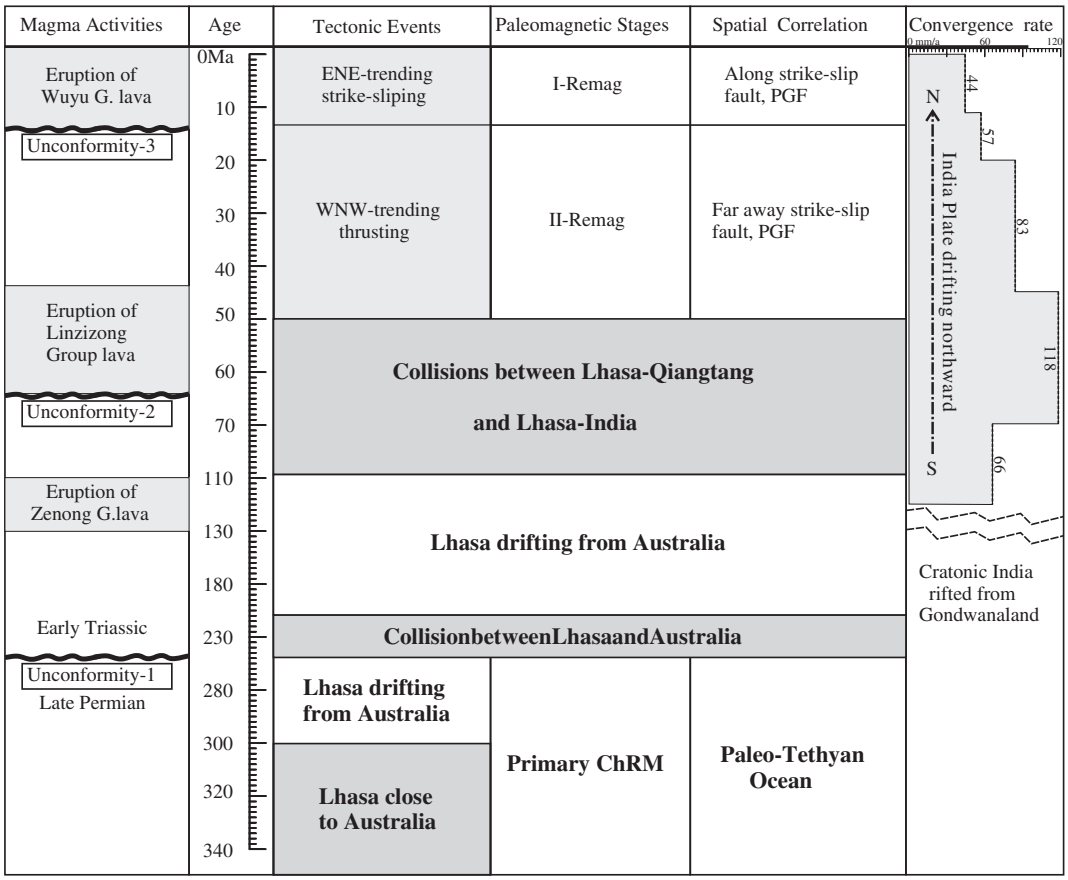


Fig. 11. Spatial and temporal relationships between controlling factors and remagnetization stages in the Xainza area, and between the Lhasa Terrane and adjacent continents. Data of three magma activities from Wang et al. (2003), Mo et al. (2007), Zhu et al. (2008); of three unconformities (1, 2, 3) from BGMRX (1993, 1997), Wang et al. (2003), Qu et al. (2003), Liu et al. (2003) and Zhu et al. (2010); of the tectonic events from Wu et al. (2003), Kapp et al. (2007a), Coleman and Hodges (1995), Yin et al. (1999), Blisniuk et al. (2001), and Taylor and Yin (2009); of the convergence rate of the Indian continent from Molnar and Stock (2009). G.: Group.

late Oligocene – early Miocene (Fig. 1; e.g., Wu et al., 2003; Wang et al., 2003; Qu et al., 2003; Kapp et al., 2007a, 2007b), (2) the maximum crustal shortening developed during the injection of the weak even partially molten India crust material into the middle crust of the convergence area in the middle–late Paleogene (e.g., Kapp et al., 2005; Yin, 2006; Chen et al., 2010), which would facilitate fluid flow supported by the calcite vein extensive in the limestones, Xiala Formation (Fig. 4E).

5.2. Late Paleozoic paleomagnetic results of the Lhasa Terrane

The primary magnetization in the rocks has generally been overprinted by multiphase deformation events in the orogen (e.g., Halim et al., 1998; Elmore et al., 2001; Zhao and Coe, 2007; Torsvik et al., 2009). In particular, the remagnetization is most likely occurred in the Lhasa Terrane because the Permian primary ChRMs of the Lhasa Terrane experienced the Qiangtang–Lhasa and India–Lhasa multiphase collisions (e.g., Lin and Watts, 1988; Chen et al., 1993; Halim et al., 1998). Therefore, the primary ChRMs reported previously from the Xainza area are suspected because the quality of former paleomagnetic data is lower than the lowest reliability criteria ($Q>2$) (van der Voo, 1990), only including two criteria dealing with age (# l) and pole position (#7) (Table 2; Ye and Li, 1987; Dong et al., 1990). So the lower paleolatitude of the Lhasa Terrane based on the former data is still excluded by the paleogeographic reconstructions because of the lower quality (e.g., Yin and Harrison, 2000; Zhu et al., 2011a).

Above all the discussion, there are two main chemical remagnetization events in the study area controlled by (1) the ENE-trending strike-slip faults in the Neogene, which only was limited into the surrouding of the faults and caused the remagnetization of the magnetite in the limestone, dolomite and silicalite; (2) the WNW-trending thrusting faults before the Miocene, which may cause the extensive remagnetization event in the limestone during the maximum crustal shortening, consistent with the Paleogene remagnetization in the chemical remagnetization in the Yanshiping area, the Qiangtang Terrane (e.g., Lin and Watts, 1988), however this remagnetization event could not affect the primary ChRMs of the detrital origin preserved in the Upper Carboniferous–Lower Permian sandstones through the tectonic fluids. Then the structure (# 5) in the study area could be demonstrated by the above remagnetization events so that the paleomagnetic quality could satisfy minimum standards of reliability for constructing APWP (i.e., $Q>2$) (Table 2; van der Voo, 1990; McElhinny and McFadden, 2000).

The former available late Carboniferous and early Permian paleomagnetic data from the Xainza area, Lhasa Terrane are listed in Table 2, and the corresponding paleopoles are shown in Fig. 9b. Four paleomagnetic studies of late Carboniferous and early Permian rocks are available within the Lhasa Terrane with $Q=3$ (Fig. 10b and Table 2). In the Xainza area, Ye and Li (1987) proposed a paleopole of 16.3° N, 320.4° E for the Lhasa Terrane in the Yongzhu Formation, late Carboniferous, but lacking of the detail number of site/sample. Dong et al. (1990) reported two paleomagnetic results of the Yongzhu Formation and the Laga Formation, although the number of sites and samples is too small (Table 2). Guo (2009) produced two late Paleozoic paleomagnetic datasets of the Yongzhu Formation and the Angjie Formation with the positive fold test (Table 2). As illustrated in Fig. 10b, we find that our new early Permian paleopole is statistically distinguishable

Table 2
Locality paleomagnetic poles from this study compared to expected directions from adjacent area in the Lhasa Terrane (reference point: 30.9° N, 88.9° E) and the Eurasian Plate and cratonic Australia.

Age(Ma)	R	N	Lat.	Lon.	Plat.	Plon.	*A95*	Q	Reference
60–65	LA	15	29.9	91.2	66.0	284.9	8.5	7	Chen et al. (2010)
~55	LA	14	30.1	90.9	73.6	274.3	7.3	6	Sun et al. (2010)
114.2	LA	15	30.5	90.1	66.4	220.3	6.9	5	Sun et al. (2008)
II-LP2x1,2,3	LI	3	30.9	88.9	76.4	158.4	15.2		This study
P1	SA	2	30.9	89.2	27.2	223.1	10.2	4	This study
P1	SA	3	30.9	89.3	27.1	218.8	9.9	3	Guo (2009)
C2	SA	5	30.9	89.3	21.9	308.2	17	3	Guo (2009)
P1	SA	5	30.9	89.2	24.5	219.3	7.8	5+1	This study + Guo (2009)
C2-P1	SA	(8)	31.0	88.7	41.8	288.7	7.0	2+1	Dong et al. (1990)
C2	SA	(10)	31.0	88.7	32.2	294.0	10.8	2+1	Dong et al. (1990)
C2	SA	?	31.0	88.7	16.3	320.4	9.1	2+1	Ye and Li (1987)
K1 (120Ma)			Eurasia		78.4	196.5	2.6		Torsvik et al. (2008)
K2 (80Ma)			Eurasia		79.6	170.0	2.6		
C2	(A)		Australia		32.1	299.5	11.9	5	Chen et al. (1994)
C2	(B)		Australia		33.8	301.2	19.2	4	Li et al. (1989)
C2	(C)		Australia		30.0	318.0	24.0	4	Camacho et al. (1991)
C2-P1	(D)		Australia		45.9	329.5	12.8	5	Idnurm and Heinrich (1993)

C2: late Carboniferous; P1: early Permian; P2: middle Permian; K1: early Cretaceous; K2: late Cretaceous; II: II–Remag; N: number of sites accepted, number showing in the parentheses indicate number of samples accepted; R: type of sampling rocks; LA: lava; LI: limestone; SA: sandstone; Lat./Lon.: latitude/longitude of sampling sites; Plat./Plon.: latitude/longitude of paleopole; *A95*: the radius that the mean pole lies within 95% confidence; Q: quality classification according to van der Voo (1990).

(at 95% confidence level) from the late Carboniferous paleopole of the Xainza locality in the former data but consistent with the early Permian paleopole of Guo (2009). Because the early Permian results (this study and Guo's) are not statistically distinguishable and because they are from the same age strata (the Angjie Formation) and areas, we have combined the site mean directions from both studies to provide a more robust paleomagnetic pole. The combined final average palaeomagnetic direction for early Permian sedimentary rocks is $D=37.9°$, $I=1.3°$, $\kappa=19.1$, $\alpha95=18.0°$ *in-situ*, and $D=46.6°$, $I=-31.2°$, $\kappa=69.0$, $\alpha95=9.3°$ after tilt correction, N=5 sites, corresponding to a pole at 24.5° N, 219.3° E with A95=7.8°. The fold tests are positive at the 95% confidence level (McFadden, 1990) and the progressive unfolding process (Watson and Enkin, 1993) (Table 1), indicating a pre-folding origin for the characteristic magnetization. Taken together with spatial and rocks limitation of the two remagnetization events and the folding age before the late Permian, the positive fold test, high-temperature ChRM stability, and precise determination of sample ChRM directions suggest a primary origin for the characteristic magnetization.

5.3. Paleogeographic implications

Because the quality of the former data meets the lowest criteria with Q>2 after the relationships between structure and remagnetization have been clarified (Table 2), these late Carboniferous paleopoles could be used to construct of the Lhasa Terrane. Contrasting with the reference poles of the cratonic Australian APWP with Q>2 (e.g., Li et al., 1989; Camacho et al., 1991; Idnurm and Heinrich, 1993; Chen et al., 1994), the paleomagentic results of the late Carboniferous sedimentary rocks from the Xainza area imply there is a little difference and the Lhasa Terrane should be the part of the northern margin of the Australia continent (Fig. 10b and c). However, there are statistically distinguishable between the Lhasa Terrane and Gondwana continents (cratonic India and Australia) in the Permian (Fig. 10b), which show not only significant relative north–south displacement ($31.9°\pm6.0°$ and $45.0°\pm6.0°$ respectively compared to the cratonic India and Australia) but also significant rotation since the early Permian (Fig. 10b). These evidences indicate the Lhasa Terrane was drifted from the cratonic Australia since the early Permian. Compared the early Permian and late Carboniferous poles of the Lhasa Terrane, there are not visible displacement but statistically clockwise rotation range from $100.3°\pm9.4°$ to $61.6°\pm8.0°$ respective compared with the late Carboniferous poles of the Ye and Li (1987) and Dong et al. (1990) (Fig. 10b and c).

This study reconstructs the paleogeography of the Lhasa Terrane prior to drift from stable Australia based on the paleomagnetic data with biogeographical and other data. Zhu et al. (2010 and unpublished data) predicts that the ultimate isolation of the Lhasa Terrane within the Paleo-Tethyan Ocean will occur during the Carboniferous–middle Permian following the reconstructions of the back-arc systems in the eastern peri-Gondwanan terranes (Murphy et al., 2011) based on the identification of the Songdo eclogite (Li et al., 2009; Yang et al., 2009) and of the Pikang granite (Zhu et al., 2009a). The paleopoles between the Lhasa Terrane and cratonic Australia are little different, indicating that the two terranes should be uniform (Fig. 10c). This interpretation is favored by the following observations: (1) the rifting and sediment accommodation of the marine rift basins in the western Australia started from 312 Ma in the Carnarvo and Bonaparte basins to 298 Ma in the Collie and Perth basins (Eyles et al., 2002), which is consistent with the climax rift stage of the Coqen basin in the Yongzhu Formation stage, late Carboniferous (Zhu et al., 2004); (2) cold water corals, dropstones (Fig. 4F) and diamictites in the Yongzhu Formation in the Lhasa Terrane indicate the cold climate similar with the eastern Gondwana in the late Carboniferous (e.g., Jin, 2002; Eyles et al., 2006; Yao et al., 2007), however, the marine faunas of the Xiala Formation, the middle Permian exhibit warm Tethyan Realm characteristics (e.g., Jin, 2002; Zheng et al., 2005; Yao et al., 2007); (3) U-Pb age and Hf-isotope analyses of detrital zircons from late Carboniferous sedimentary rocks in the Lhasa Terrane supported the close affinity between the Lhasa Terrane and cratonic Australia (Zhu et al., 2011a); (4) the coeval Precambrian metamorphism (~720 Ma) occurred in the Lhasa Terrane (Zhang et al., 2010) and northern Australia (e.g., Bagas, 2004) support the paleogeographic connection between these two regions. Based on the paleomagnetic data reported here and previous works, we argue that the Lhasa terrane was profoundly separated from the east Gondwanaland during this period while the Carboniferous paleomagnetic results indicating a connection between the Lhasa and Australia, therefore, the new early Permian paleopole indicates that the Lhasa terrane should have drifted from Australia of east Gondwanaland by the early Permian.

6. Conclusions

The Angjie Formation detrital rocks reveal well-defined remanence components that are carried by hematite. The positive fold test suggests a primary magnetization that was acquired during the early Permian. Futhermore the Angjie Formation pole is obvious not consistent with the Australian and Indian poles in the Permian but the Carboniferous pole in the Lhasa Terrane overlaps with the Australian poles from 300 to 350 Ma. The lower paleolatitude in the early Permian also indicates the Lhasa Terrane should have rifted from the northern margin of Australia after the Carboniferous.

7. Note added in proof

The interpretation and conclusion that "the paleomagnetic results of the late Carboniferous sedimentary rocks from the Xainza area imply there is a little difference between Lhasa and Australia in paleolatitudes and the Lhasa Terrane should be the part of the northern margin of the Australia continent" in this article is not supported by the data and thus is in error. The overlap of the Carboniferous paleopoles for the Lhasa terrane and Australia was just a coincidence, not implying that Lhasa and Australia were next to each other in Carboniferous times. We thank Dr. Peter Lippert for pointing out this error.

Acknowledgements

We are grateful to criticisms of two journal anonymous reviewers and the editor. We thank Drivers Hongming Luo and Wang Aa, Dr. Yushuai Wei and Licheng Wang for their assistance in the field; Dr. Zhenyu Li and Chengying Liu for help with Paleomagnetic experiments; Dr. Cuihua Chen for help with polished thins experiment. We are grateful for helpful discussions with Dr. Dicheng Zhu, Xiumian Hu and Junshan Chen. This study benefited from financial support by the National Key Project for Basic Research of China (Project 2006CB701400), New Teacher Fund for Doctor Station, the Ministry of Education (Project 20105122120013), Educational Commission of Sichuan Province of China (Project 09ZA005), Construction Project of National Key Subject (Cultivating) in Mineralogy, Petrology, and Ore Deposit (Project SZD0407), China Geological Survey (Project 1212010818095).

References

Acton, G.D., 1999. Apparent polar wander of India since the Cretaceous with implications for regional tectonics and true polar wander. In: Radhakrishna, T., and Piper, J.D.A. (Eds.), The Indian Subcontinent and Gondwana: A Palaeomagnetic and Rock Magnetic Perspective. Memoir Geological Society of India 44, pp. 129–175.

Aitchison, J.C., Davis, A.M., 2004. Evidence for the multiphase nature of the India–Asia collision from the Yarlung Tsangpo suture zone, Tibet. In: Malpas, J., Fletcher, C.J.N., Ali, J.R., Aitchison, J.C. (Eds.), Aspects of the tectonic evolution of China: Geological Society, London, Special Publications, 226, pp. 217–233.

Aitchison, J., Xia, X., Baxter, A.T., Ali, J.R., 2011. Detrital zircon U–Pb ages along the Yarlung-Tsangpo suture zone, Tibet: implications for oblique convergence and collision between India and Asia. Gondwana Research 20, 691–709.

Allégre, C.J., Courtillot, V., Tapponnier, P., Hirn, A., Mattauer, M., Coulon, C., Jaeger, J.J., Achache, J., Schärer, U., Marcoux, J., Burg, J.P., Girardeau, J., Armijo, R., Gariépy, C., Göpel, C., Li, T.D., Xiao, X.C., Chang, C.F., Li, G.Q., Lin, B.Y., Teng, J.W., Wang, N.W., Chen, G.M., Han, T.L., Wang, X.B., Den, W.M., Sheng, H.B., Cao, Y.G., Zhou, J., Qiu, H.R., Bao, P.S., Wang, S.C., Wang, B.X., Zhou, Y.X., Xu, R.H., 1984. Structure and evolution of the Himalayan–Tibet orogenic belt. Nature 307, 17–22.

Bagas, L., 2004. Proterozoic evolution and tectonic setting of the northwest Paterson Orogen, Western Australia. Precambrian Research 128, 475–496.

BGMRX (Bureau of Geology Mineral Resources of Xizang Autonomous Region), 1993. Regional Geology of Xizang (Tibet) Autonomous Region. Geological Publishing House, Beijing, pp. 1–707 (in Chinese with English abstract).

BGMRX (Bureau of Geology Mineral Resources of Xizang Autonomous Region), 1997. Stratigraphy (Lithostratigraphic) of Xizang Autonomous Region. China University of Geoscienecs Press, Wuhan, pp. 145–162 (in Chinese with English abstract).

Blisniuk, P.M., Hacker, B.R., Glodny, J., Ratschbacher, L., Siwen, B., Zhenhun, W., McWilliams, M.O., Calvert, A., 2001. Normal faulting in central Tibet since at least 13.5 Myr ago. Nature 412, 628–632.

Camacho, A., Symons, B., Schmidt, P.W., 1991. Geological and palaeomagnetic significance of the Kulgera Dyke Swarm, Musgrave Block, NT, Australia. Geophysical Journal International 107, 37–45.

Chen, J.S., Huang, B.C., Sun, L.S., 2010. New constraints to the onset of the India–Asia collision: paleomagnetic reconnaissance on the Linzizong Group in the Lhasa Block, China. Tectonophysics 489, 189–209.

Chen, Y., Cogne, J.P., Courtillot, V., Tapponnier, P., Zhu, Y.X., 1993. Cretaceous paleomagnetic results from western Tibet and tectonic implications. Journal of Geophysical Research 98. doi:10.1029/93JB01006.

Chen, Z., Li, Z.X., Powell, C.M., Balme, B.E., 1994. An Early Carboniferous paleomagnetic pole for Gondwanaland: new results from the Mount Eclipse Sandstone in the Ngalia Basin, central Australia. Journal of Geophysical Research 99, 2909–2924.

Chen, W., Yang, T., Zhang, S., Yang, Z., Li, H., Wu, H., Zhang, J., Ma, Y., Cai, F., 2011. Paleomagnetic results from the Early Cretaceous Zenong Group volcanic rocks, Cuoqin, Tibet, and their paleogeographic implications. Gondwana Research. doi:10.1016/j.gr.2011.07.019.

Chung, S.L., Chu, M.F., Zhang, Y.Z., Xie, Y.W., Lo, C.H., Lee, T.Y., Lan, C.Y., Li, X.H., Zhang, Q., Wang, Y.Z., 2005. Tibetan tectonic evolution inferred from spatial and temporal variations in post-collisional magmatism. Earth-Science Reviews 68, 173–196.

Coe, R.S., Stock, G.M., Lyons, J.J., Beitler, B., Bowen, G.J., 2005. Yellowstone hotspot volcanism in California? A paleomagnetic test of the Lovejoy flood basalt hypothesis. Geology 33, 697–700.

Coleman, M., Hodges, K., 1995. Evidence for Tibetan plateau uplift before 14 Myr ago from a new minimum age for east–west extension. Nature 374, 49–52.

Dewey, J.F., Shackelton, R.M., Chang, C., Sun, Y., 1988. The tectonic evolution of the Tibetan Plateau. Philosophical Transactions of the Royal Society of London A327, 379–413.

Dong, X.B., Wang, Z.M., Tan, C.Z., Yang, H.X., Cheng, L.R., Zhou, Y.X., 1990. New results of paleomagnetic studies of the Qinghai–Tibetan plateau. Geological Review 37, 160–164 (in Chinese with English abstract).

Dupont-Nivet, G., Lippert, P.C., van Hinsbergen, D.J.J., Meijers, M.J.M., Kapp, P., 2010. Palaeolatitude and age of the Indo-Asia collision: palaeomagnetic constraints. Geophysical Journal International 182, 1189–1198.

Elmore, R.D., Kelley, J., Evans, M., Lewchuk, M.T., 2001. Remagnetization and orogenic fluids: testing the hypothesis in the central Appalachians. Geophysical Journal International 144, 568–576.

Enkin, R.J., 2003. The direction-correction tilt test: an all-purpose tilt/fold test for paleomagnetic studies. Earth and Planetary Science Letters 212, 151–166.

Enkin, R.J., Baker, J., Evenchick, C.A., 2007. Diagenetic remagnetization in Bowser Basin: implications for timing and tectonics. Bulletin of Canadian Petroleum Geology 55, 306–319.

Eyles, N., Mory, A.J., Backhouse, J., 2002. Carboniferous–Permian palynostratigraphy of west Australian marine rift basins: resolving tectonic and eustatic controls during Gondwanan glaciations. Palaeogeography, Palaeoclimatology, Palaeoecology 184, 305–319.

Eyles, N., Mory, A.J., Eyles, C.H., 2006. 50-million-year-long record of glacial to postglacial marine environments preserved in a Carboniferous-Lower Permian Graben, Northern Perth Basin, Western Australia. Journal of Sedimentary Research 76, 618–632.

Fisher, R.A., 1953. Dispersion on a sphere. Proceedings of the Royal Society A217, 295–305.

Golonka, J., Ford, D., 2000. Pangean (Late Carboniferous–Middle Jurassic) paleoenvironment and lithofacies. Palaeogeography, Palaeoclimatology, Palaeoecology 161, 1–34.

Guo, Q., 2009. Research on Paleomagnetism of Gangdise Plate of Qinghai-Tibet Plateau in Late Paleozoic. Northwest Universtiy Master's Dissertation, pp. 1–60 (in Chinese with English abstract).

Halim, N., Cogne, J.P., Chen, Y., Atasiei, R., Besse, J., Courtillot, V., Gilder, S., Marcoux, J., Zhao, R.L., 1998. New Cretaceous and Early Tertiary paleomagnetic results from Xinin–Lanzhou basin, Kunlun and Qiangtang blocks, China: implications on the geodynamic evolution of Asia. Journal of Geophysical Research 103. doi:10.1029/98JB01118.

Idnurm, M., Heinrich, C.A., 1993. A palaeomagnetic study of hydrothermal activity and uranium mineralization at Mt Painter, South Australia. Australian Journal of Earth Sciences 40, 87–101.

Jin, X.C., 2002. Permo-Carboniferous sequences of Gondwana affinity in southwest China and their paleogeographic implications. Journal of Asian Earth Sciences 6, 633–646.

Kapp, P., DeCelles, P.G., Gehrels, G.E., Heizler, M., Ding, L., 2007a. Geological records of the Lhasa–Qiangtang and Indo-Asian collisions in the Nima area of central Tibet. Geological Society of America Bulletin 119, 917–932.

Kapp, P., DeCelles, P.G., Leier, A.L., Fabijanic, J.M., He, S., Pullen, A., Gehrels, G.E., Ding, L., 2007b. The Gangdese retroarc thrust belt revealed. Geological Society of America Today 17, 4–9.

Kapp, P., Yin, A., Harrison, T.M., Ding, L., 2005. Cretaceous–Tertiary shortening, basin development, and volcanism in central Tibet. Geological Society of America Bulletin 117, 865–878.

Katz, B., Elmore, R.D., Engel, M.H., 1998. Authigenesis of magnetite in organic-rich sediment next to a dike: implications for thermoviscous and chemical remagnetizations. Earth and Planetary Science Letters 163, 221–234.

Kirschvink, J.L., 1980. The least-square line and plane and the analysis of paleomagnetic data. Geophysical Journal of the Royal Astronomical Society 62, 699–718.

Li, H.Q., Xu, Z.Q., Yang, J.S., Cai, Z.H., Chen, S.Y., Tang, Z.M., 2009. Records of Indosinian Orogenesis in Lhasa Terrane, Tibet. Journal of Earth Science 20, 348–363.

Li, P.W., Gao, R., Cui, J.W., Guan, Y., 2004. Paleomagnetic analysis of eastern Tibet: implications for the collisional and amalgamation history of the Three Rivers Region, SW China. Journal of Asian Earth Sciences 24, 291–310.

Li, Z.X., Powell, C.M., Schmidt, P.W., 1989. Syn-deformational remanent magnetization of the Mount Eclipse Sandstone, central Australia. Geophysical Journal International 99, 205–222.
Lin, J.L., Watts, D.R., 1988. Paleomagnetic results from the Tibetan Plateau. Philosophical Transactions of the Royal Society A327, 239–262.
Liu, D.Z., Tao, X.F., Ma, R.Z., Shi, H., Zhu, L.D., Hu, X.W., 2003. 1:250,000 geological report of the Coqen region with geological map, Tibet. Chengdu University of Technology Geological Survey, Chengdu.
Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock coercivity and unblocking temperature properties. Geophysical Research Letters 17, 159–162.
McElhinny, M., McFadden, P., 2000. Paleomagnetism: Continents and Oceans. Academic Press, California, pp. 1–394.
McElhinny, M.W., 1964. Statistical significance of the fold test in paleomagnetism. Geophysical Journal International 8, 338–340.
McFadden, P.L., 1990. A new fold test for paleomagnetic studies. Geophysical Journal International 103, 163–169.
Metcalfe, I., 2006. Palaeozoic and Mesozoic tectonic evolution and palaeogeography of East Asian crustal fragments: The Korean Peninsula in context. Gondwana Research 9, 24–46.
Mo, X.X., Hou, Z.Q., Niu, Y.L., Dong, G.C., Qu, X.M., Zhao, Z.D., Yang, Z.M., 2007. Mantle contributions to crustal thickening during continental collision: Evidence from Cenozoic igneous rocks in southern Tibet. Lithos 96, 225–242.
Molnar, P., Stock, J.M., 2009. Slowing of India's convergence with Eurasia since 20 Ma and its implications for Tibetan mantle dynamics. Tectoncis 28, TC3001. doi:10.1029/2008TC002271.
Murphy, J.B., van Staal, C.R., Collins, W.J., 2011. A comparison of the evolution of arc complexes in Paleozoic interior and peripheral orogens: Speculations on geodynamic correlations. Gondwana Research 19, 812–827.
Najman, Y., Appel, E., Boudagher-Fadel, M., Bown, P., Carter, A., Garzanti, E., Godin, L., Han, J., Liebke, U., Oliver, G., Parrish, R., Vezzoli, G., 2010. The timing of India-Asia collision: geological, biostratigraphic and palaeomagnetic constraints. Journal of Geophysical Research 115. doi:10.1029/2010JB007673.
Oliver, J., 1986. Fluids expelled tectonically from orogenic belts: their role in hydrocarbon migration and other geologic phenomena. Geology 14, 99–102.
Park, Y.H., Doh, S.J., Suk, D., 2003. Chemical remagnetization of the Upper Carboniferous-Lower Triassic Pyeongan Supergroup in the Jeongseon area, Korea: fluid migration through the Ogcheon Fold Belt. Geophysical Journal International 154, 89–103.
Qu, Y.G., Wang, Y.S., Duan, J.X., Zhang, S.Q., Wang, Z.H., Lv, P., Liu, G.Z., Li, X.B., Feng, D.C., Xie, Y.H., Yu, X.W., Jiang, X.F., Sun, Z.G., Li, Q.W., Liang, S.F., Wang, H.S., Guo, S.S., 2003. 1:250,000 geological report of the Duoba region with geological map, Tibet. Jinlin Institute of Geological Survey, Changchun.
Scotese, C.R., Boucot, A.J., McKerrow, W.S., 1999. Gondwanan palaeogeography and palaeoclimatology. Journal of African Earth Sciences 28, 99–114.
Şengör, A.M.C., 1979. Mid-Mesozoic closure of Permo-Triassic Tethys and its implications. Nature 279, 590–593.
Sun, Z.M., Jiang, W., Li, H.B., Pei, J.L., Zhu, Z.M., 2010. New paleomagnetic results of Paleocene volcanic rocks from the Lhasa block: Tectonic implications for the collision of India and Asia. Tectonophysics 490, 257–266.
Sun, Z., Pei, J., Li, H., Wu, W., Jiang, W., Zhu, Z., Wang, X., Yang, Z., 2011. Palaeomagnetism of the late Cretaceous sediments from southern Tibet: Evidence for the consistent paleolatitudes of the southern margin of the Eurasia prior to the collision with India. Gondwana Research. doi:10.1016/j.gr.2011.08.003.
Sun, Z.M., Jiang, W., Pei, J.L., Li, H.B., 2008. New Early Cretaceous paleomagnetic data from volcanic of the eastern Lhasa Block and its tectonic implications. Acta Petrologica Sinica 24, 1621–1626 (in Chinese with English abstract).
Taylor, M., Yin, A., 2009. Active structures of the Himalayan-Tibetan orogen and their relationships to earthquake distribution, contemporary strain field, and Cenozoic volcanism. Geosphere 5, 199–214.
Torsvik, T.H., Müller, R.D., Van der Voo, R., Steinberger, B., Gaina, C., 2008. Global plate motion frames: toward a unified model. Reviews of Geophysics 46, 1–44.
Torsvik, T.H., Paulsen, T.S., Hughes, N.C., Myrow, P.M., GanerØd, M., 2009. The Tethyan Himalaya: palaeogeographical and tectonic constraints from Ordovician palaeomagnetic data. Journal of the Geological Society of London 166, 679–687.
van der Voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics 180, 1–9.
Wang, C.S., Zhao, X.X., Liu, Z.F., Lippert, P.C., Graham, S.A., Coe, R.S., Yi, H.S., Zhu, L.D., Liu, S., Li, Y.L., 2008. Constraints on the early uplift history of the Tibetan Plateau. Proceedings of the National Academy of Sciences of the United States of America 105, 4987–4992.
Wang, T.W., Cheng, L.R., Li, C., Wu, S.Z., Zhao, J.C., He, Z.H., Zhang, Y.J., Zhu, Z.Y., Yang, D.M., 2003. 1:250,000 geological report of the Xainza region with geological map, Tibet. Jilin University Geological Survey, Changchun.
Watson, G.S., Enkin, R.J., 1993. The fold test in paleomagnetism as a parameter estimation problem. Geophysical Research Letters 20, 2135–2137.
Wu, Z.H., Barosh, P.J., Wu, Z.H., Hu, D.G., Zhao, X., Ye, P.S., 2008. Vast early Miocene lakes of the central Tibetan Plateau. Geological Society of America Bulletin 120, 1326–1337.
Wu, Z.H., Ye, P.S., Hu, D.G., Liu, Q.S., 2003. Thrust System of the North Lhasa Block. Geological Review 49, 74–81 (in Chinese with English abstract).
Yang, J.S., Xu, Z.Q., Li, Z.L., Xu, X.Z., Li, T.F., Ren, Y.F., Li, H.Q., Chen, S.Y., Robinson, P.T., 2009. Discovery of an eclogite belt in the Lhasa block, Tibet: A new border for Paleo-Tethys? Journal of Asian Earth Sciences 34, 76–89.
Yao, J.X., Ji, Z.S., Wu, G.C., Zhan, L.P., Liu, G.Z., Jiang, Z.T., Fu, Y.H., 2007. Deri'angma-Xiala section in the Xainza area, Tibet, China: A bridge for the stratigraphic and paleontological correlation between Gondwana and Tethys during the Late Carboniferous and Early Permian. Geological Bulletin of China 26, 31–41 (in Chinese with English abstract).
Ye, X.H., Li, J.F., 1987. Palaeomagnetism and evolution of Tibet plates and Tethys. Journal of Chengdu College of Geology 14, 65–79 (in Chinese with English abstract).
Yin, A., 2006. Cenozoic tectonic evolution of the Himalayan orogen as constrained by along-strike variation of structural geometry, exhumation history, and foreland sedimentation. Earth-Science Reviews 76, 1–131.
Yin, A., Harrison, T.M., 2000. Geologic evolution of the Himalayan–Tibetan orogen. Annual Review of Earth and Planetary Sciences 28, 211–280.
Yin, A., Kapp, P., Murphy, M., Manning, C., Harrison, T.M., Grove, M., Ding, L., Deng, X., Wu, C., 1999. Significant late Neogene east–west extension in northern Tibet. Geology 27, 787–790.
Zhang, Z.M., Dong, X., Geng, G.S., Wang, W., Yu, F., Liu, F., 2010. Precambrian metamorphism of the northern Lhasa Terrane, south Tibet and its tectonic implications. Acta Geologica Sinica 84, 449–456 (in Chinese with English abstract).
Zhao, X.X., Coe, R.S., 2007. Plate tectonics of China. In: Gubbins, D., Herrero-Bervera, E. (Eds.), Encyclopedia of Geomagnetism and Paleomagnetism: Encyclopedia of Earth Sciences Series, XXXVI. Kluwer, Amsterdam, Netherlands, pp. 816–828.
Zheng, C.Z., Wang, Y.S., Zhang, S.Q., 2005. The Carboniferous-Permian biostratigraphic division of Deriangmato-Xialashan of the Xainza area, Northern Tibet. Journal of Stratigraphy 29, 520–528 (in Chinese with English abstract).
Zhu, D.C., Mo, X.X., Niu, Y.L., Zhao, Z.D., Wang, L.Q., Liu, Y.S., Wu, F.Y., 2009a. Geochemical investigation of Early Cretaceous igneous rocks along an east–west traverse throughout the central Lhasa Terrane, Tibet. Chemical Geology 268, 298–312.
Zhu, D.C., Mo, X.X., Niu, Y.L., Zhao, Z.D., Wang, L.Q., Pan, G.T., Wu, F.Y., 2009b. Zircon U-Pb dating and in-situ Hf isotopic analysis of Permian peraluminous granite in the Lhasa terrane, southern Tibet: Implications for Permian collisional orogeny and paleogeography. Tectonophysics 469, 48–60.
Zhu, D.C., Mo, X.X., Zhao, Z.D., Niu, Y.L., Wang, L.Q., Chu, Q.H., Pan, G.T., Xu, J.F., Zhou, C.Y., 2010. Presence of Permian extension- and arc-type magmatism in southern Tibet: Paleogeographic implications. Geological Society of America Bulletin 122, 979–993.
Zhu, D.C., Mo, X.X., Zhao, Z.D., Xu, J.F., Wang, L.Q., Chen, H.H., Dong, G.C., Zhou, C.Y., Zhou, S., 2008. Zircon U-Pb geochronology of Zenong Group volcanic rocks in Coqen area of the Gangdese, Tibet and tectonic significance. Acta Petrologica Sinica 24, 401–412 (in Chinese with English abstract).
Zhu, D.C., Zhao, Z.D., Niu, Y.L., Dilek, Y., Mo, X.X., 2011a. Lhasa terrane in southern Tibet came from Australia. Geology 39, 727–730.
Zhu, D.C., Zhao, Z.D., Niu, Y.L., Mo, X.X., Chung, S.L., Hou, Z.Q., Wang, L.Q., Wu, F.Y., 2011b. The Lhasa Terrane: Record of a microcontinent and its histories of drift and growth. Earth and Planetary Science Letters 301, 241–255.
Zhu, L.D., Liu, D.Z., Tao, X.F., Shi, H., Ma, R.Z., Hu, X.W., 2004. Evolution of the lithofacies and paleogeography in the Coqen area of Tibet during the Carboniferous to early Permian. Advance in Earth Sciences 19, 46–49 (in Chinese with English abstract).
Zhu, L.D., Wang, C.S., Zheng, H.B., Xiang, F., Yi, H.S., Liu, D.Z., 2006. Tectonic and sedimentary evolution of basins in the northeast of Qinghai-Tibet Plateau and their implication for the northward growth of the Plateau. Palaeogeography, Palaeoclimatology, Palaeoecology 241, 49–60.
Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods on Paleomagnetism. Elsevier, Amsterdam, pp. 254–286.